INCH-POUND
A-A-59312D
22 December 2014
Superseding
A-A-59312C
31 July 2006

COMMERCIAL ITEM DESCRIPTION

TRACTOR, AIRCRAFT TOWING, FLIGHT LINE

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE.

This Commercial Item Description (CID) covers the requirements for the Flight Line Tow Tractor (FLTT), hereafter referred to as FLTT. The FLTT is based on a commercial truck chassis, also known as a chassis cab, which is modified for use as a FLTT in worldwide service. The FLTT is capable of towing trailers weighing up to 40,000 pounds, and military aircraft weighing up to 70,000 pounds on grades up to 2.5% on U.S. Air Force installations.

2. CLASSIFICATION.

2.1 FLTT classes. The two classes specified shall be of the same design, with the only differences being the components necessary for a four-wheel drive FLTT. The FLTT shall be one of the following classes specified:

Class 1 – two-wheel drive FLTT

Class 2 – four-wheel drive FLTT

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/WNZEB, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1740

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS.

3.1 FLTT description. The FLTT is based on a commercial truck chassis, which is modified for use as a FLTT in worldwide service. The Air Force (AF) uses the FLTT to tow Aircraft Ground Equipment (AGE), munitions trailers, and military aircraft. The modifications to the commercial truck chassis include shortening the length of the chassis by cutting the frame rails, and shortening the wheelbase by relocating the rear axle, suspension, and other affected components forward on the original chassis frame rails. Other modifications include the addition of a ballast bed over the rear axle for traction, a tool box, and hitches for towing.

3.2 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. The FLTT shall be designed and manufactured in accordance with Society of Automotive Engineers (SAE) specifications ARP1247 and AIR 1363. The FLTT shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. The FLTT shall have an all-welded, one-piece frame structure.

3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted.

3.2.1.2 Finish. The exterior finish color of the FLTT shall be the manufacturer's standard commercial paint color. Prior to the application of the finish top coat, any cab or chassis component or assembly specifically manufactured by or for the FLTT manufacturer, from sheet, plate, angular, or tubular steel, shall be coated with a zinc-rich primer. The top surfaces of the FLTT that are stepped on in association with routine maintenance shall be covered with anti-skid material. Prior to applying the finish top coat, the chassis cab's OEM finish shall be properly prepared by surface conditioning (scuffing with an abrasive material and cleaning) the paint at a minimum. The finish top coat shall extend into the door jambs and hood jambs to the extent that the OEM finish color cannot be seen through the gaps between the body panels when the doors and hood are closed.

3.2.1.3 Exclusion of water. The design of the FLTT shall be such as to prevent water leaking into, or being driven into, any part of the FLTT interior when either in an operating or travelling configuration. All windows, doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the FLTT that could collect and retain water.

3.2.1.3.1 Fluid traps and faying surfaces. There shall be no fluid traps on the FLTT. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.3.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.1.3.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall permit drainage of all fluids when the FLTT is on a 12% longitudinal grade facing both up and down and on a 10% side slope facing both directions. The minimum size of the drain holes shall be 0.25 inch.

3.2.1.4 Corrosion protection. The FLTT shall be corrosion-protected with a thick, hard, black, commercially-available Corrosion Preventive Compound (CPC) that is difficult to remove (CORTEC VpCI 375 or equivalent). Application shall include the exposed chassis-frame metal underneath the FLTT (excluding the drive train), the inside of the wheel wells, and behind the fender skirts. Any equivalents used shall meet the below salient characteristics in Table I when applied on carbon steel 1010 Q-panels:

TABLE I. VpCI 375 equivalent salient characteristics.

Specification	Dry Film Thickness (Mils)	Hours to Failure
ASTM B117	4.0-4.5	2500+
ASTM D1748	4.0-4.5	2000+
ASTM D870	4.0	312+

In addition to the equivalent requirements in Table I, the pencil hardness in accordance with American Society for Testing and Materials (ASTM) D3363 shall be “H”, the temperature resistance shall be -150 ^{o}F to 350 ^{o}F, and the adhesion in accordance with ASTM D3359 shall be “5B”. When specified (see 7.2 c), FLTTs that are to be shipped overseas shall be temporarily corrosion-protected from the effect of salt spray both on the ship and at port with the application of a thin, soft-film, semi-transparent, commercially-available CPC that is easily removable by washing or with a low pressure steam (Cortec VpCI 389 or equivalent). Application of the temporary CPC shall be to the exterior of the FLTT, in critical or corrosion sensitive areas of the FLTT. Any equivalents used shall meet the below salient characteristics in Table II when applied on SAE 1010 carbon steel:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

TABLE II. VpCI 389 equivalent salient characteristics.

Specification	Dry Film Thickness (Mils)	Hours to Failure
ASTM B117	2.0	600+
ASTM D1748	1.0	1000+
ASTM G 85-94	1.0	600+

3.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of FED-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of FED-STD-595. The center of gravity of the FLTT shall be stenciled on the unit within 1.0 inch of the calculated center of gravity. Tire inflation pressures shall be marked above all wheel well openings, with ½" high letters/numbers that are black in color. A decal shall be adhered near the top of each wheel well opening that warns personnel of the dangers of placing any part of the body into the wheel well area. A decal shall be included on the front and rear of the vehicle near the pintle hook position that states: MAXIMUM TOWING CAPACITY: 4,500 LBS DRAWBAR PULL FORCE / 40,000 LBS GVW TRAILERS / 70,000 LBS GVW AIRCRAFT.

3.2.3 Identification and data plates.

3.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the FLTT in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, contract number, registration number, and National Stock Number (NSN).

3.2.3.2 Transportation data plate. The FLTT shall be provided with a permanently-marked transportation data plate constructed of a non-corroding metal, mounted on the exterior surface of the FLTT. The plate shall contain at least the following information:

a. Side and rear silhouette views of the FLTT.

b. Horizontal and vertical location of the center of gravity of the FLTT in air transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

g. Tie down information.

3.2.3.3 Lubrication data plate. The FLTT shall be provided with a permanently-marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the FLTT that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.2.3.4 OEM data plates. If any information found on the factory-original commercial truck chassis data plates (normally found in the door jambs) is inconsistent with the modified commercial truck chassis design, remove and replace them with data plates that contain updated and correct applicable information (such as rim and tire sizes, tire air pressures, GVW, etc.) that pertains to the new design.

3.2.4 Environment, Safety, and Occupational Health (ESOH).

3.2.4.1 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The FLTT shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. The FLTT shall not contain or use hexavalent chromium. For the purposes of this requirement, the Class I ODS and controlled substances identified in Chapter 4 of AFI 32-7086 shall not be used in any system, component, or process.

3.2.4.2 Component protection. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the FLTT with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.2.4.4 Fire extinguisher. The FLTT shall be equipped with two 5 pound (minimum) dry chemical type fire extinguishers in accordance with A-A-393 with mounting brackets, and a UL rating of 20B:C (minimum). Extinguishers shall be mounted on the left and right side of the FLTT, rear of the cab.

3.2.5 Electromagnetic interference (EMI). The FLTT shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 and RS103.

3.2.6 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Tapped threads shall have a minimum thread engagement in accordance with Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

TABLE III. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.2.7 Welders and welding. Welding and post heat treatment procedures shall be in accordance with the American Welding Society (AWS) welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.2.8 Special tools. The design of the item shall minimize the requirement for special tools (see 7.3.2). All special tools shall be provided with, and stored on, the FLTT.

3.2.9 Cab specifications.

3.2.9.1 Cab design. The FLTT shall be equipped with a three-man cab. Seating shall be covered in cloth. The first entrance step to the cab shall be a full-length, running board style step and shall not exceed 18 inches in height from the ground. The outer edge of this running board style step shall be offset out from the outer edge of the cab door opening by at least 4.75 inches. Cab instrumentation shall feature an hour meter in addition to the standard instrumentation. When an air system for towing trailers is specified (see 7.2 d), cab instrumentation shall include an air pressure gauge and a low-pressure indicator with buzzer and/or warning light. The FLTT shall have a fixed, covered toolbox with a hinged lid and latching mechanism located behind the cab in an easily accessible location. The toolbox lid shall be supported when opened, and the latching mechanism shall not bind when the toolbox lid is closed. The toolbox shall be a minimum of 36 inches in length, 12 inches in width, and 9 inches in height, and shall be constructed from a minimum of 11 gauge steel. The FLTT shall be equipped with an air conditioning system. The FLTT cab shall be equipped with wipers, a windshield washer, driver and passenger-door rear-view mirrors, windshield-area rear-view mirror, visors, heater/defroster, interior dome light, seat belts, and other equipment as found in standard commercial offerings. The cab shall feature a rear sliding window and a 12V power outlet (cigarette lighter type).

3.2.9.2 Cab visibility. The FLTT shall be designed to provide maximum visibility in all directions for the range of persons represented between the 5th percentile female driver and the 95th percentile male driver in accordance with Table B-IV of MIL-STD-1472. The three rear pintle hooks shall be visible either directly or via mirrors from the driver's position. If required, one mirror per pintle hook shall be provided to complete this visibility requirement. A plane defined by a line 18.00 inches above the rear pintle hooks and a point represented by the driver's eye shall not be obstructed by any structure or feature on the truck. The light bar is exempt from this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.2.9.3 Cab noise levels. Interior cab sound level shall not exceed 84 dB(A) with doors and windows closed. If the sound level is above 84 dB(A), the contractor shall provide a hazardous noise warning plate stating that operator's hearing protection is required when operating this FLTT. The plate shall be made of a corrosion resistant material and shall be at least 4.0 square inches. It shall have black lettering on a yellow background and shall be located to be visible as the operator enters the cab, or is seated. Sound level shall not exceed 94 dB(A).

3.2.10 Tires and wheels. The FLTT shall be equipped with single tubeless steel belted radial tires with on/off-road type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application.

3.2.11 Air system. When specified (see 7.2 d), the FLTT shall have an air system for towing trailers with air brakes. The air system shall be an over-the-road commercial truck/tractor design to include a reservoir tank with heated and automatic drain valves, automotive compressor, dryer, trailer protection valve, low-pressure indicator, service and emergency glad hands, glad hand covers with lanyards, and any other necessary components. The glad hands shall be in accordance with SAE J318, and shall be polarized and identified as "service" and "emergency". Components of the air system that are sensitive to the elements (rain water, road water, ice build-up, road grit, etc.) and installed in positions on the FLTT that are vulnerable to those elements shall be covered or shielded to prevent the premature failure of those components over time or during service. The configuration of the air system shall be the same on the Class 1 and Class 2 FLTT variants.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The FLTT shall be capable of operating in ambient temperatures ranging from -20 °F, or -40 ^{o}F when Winterization A is specified (see 7.2 e), to 125 ^{o}F.

3.3.2 Storage temperature range. The FLTT shall be capable of being stored in ambient temperatures ranging from -65 °F to 140°F.

3.4 Weight and dimensions.

3.4.1 Weight. The FLTT shall have a minimum gross vehicle weight (GVW) of 8,000 pounds. However, the GVW shall not exceed gross vehicle weight rating (GVWR) less 1,500 pounds. The minimum GVWR required for this application is 9,500 pounds.

3.4.2 Dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed Table IV:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

TABLE IV. Dimensions.

Length	200 inches, without pintle hooks
Width	100 inches, without mirrors
Height	100 inches

3.5 Transportability.

3.5.1 Air transportability. The FLTT shall be transportable on C-130, C-17, and C-5 aircraft in accordance with MIL-STD-1791. In all air transport configurations, the FLTT shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The FLTT shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The FLTT shall drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.5.1.1 Shoring. The use of shoring to load the aircraft (referred to as approach shoring and step-up shoring) and during flight (referred to as load shoring, parking shoring, and sleeper shoring) is permitted, but not desired. The use of shoring, especially in loading the aircraft, is not an alternative to a good FLTT design. The FLTT shall be designed to the maximum extent possible with adequate clearances for the FLTT to drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.5.1.2 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi).

3.5.1.3 Air transportable configuration. The air transportable configuration shall include all equipment, ¾ tank of fuel, and no crewmembers. At least 6 inches sidewall and 6 inches overhead clearance shall be maintained between the FLTT and the aircraft at all times during loading and flight. The restrained FLTT shall allow for loadmaster in-flight access from the front to the rear of the aircraft.

3.5.1.4 Equipment removal/reconfiguration. Preparation and restoration of the FLTT for air transportability shall take no more than 30 minutes for two persons using common non-powered hand tools. All equipment removed shall be stored on the FLTT; caps and plugs shall permit driving and storage in transport configuration.

3.5.2 Tie downs. The FLTT shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 10,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the FLTT. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.5.3 Lifting provisions. The FLTT shall be equipped with sufficient attachment points so located that it can be lifted by crane; each attachment point shall be marked "Lift Point". The lifting provisions shall be in accordance with 5.1 through 5.1.4 of MIL-STD-209.

3.6 Performance.

3.6.1 Mobility. The FLTT shall be capable of 35 miles per hour (MPH) (minimum) on flat asphalt or concrete surfaces without a trailing load. The FLTT shall be governed not to exceed 40 MPH (plus 0, minus 3), unless the rear axle reduction ratio prevents speeds greater than 40 MPH. The preferred method to control the FLTT speed is through the selection of the axle gear ratio. Reverse speed shall be a minimum of 2.5 MPH on flat surfaces. The FLTT shall be capable of towing 40,000 lb trailers from a stop to 10 MPH (minimum) and 70,000 lb aircraft from a stop to 5 MPH (minimum) on grades up to 2.5% on asphalt or concrete surfaces. The FLTT shall be capable of operating on 10 percent side slopes and 12 percent longitudinal grades without fluid spillage.

3.6.2 Drawbar pull force (DBPF). The FLTT shall have a minimum DBPF of 4,500 pounds on dry level concrete (coefficient of friction of 0.8).

3.6.3 Driveline and steering. The Class 2 FLTT shall be offered in a part-time, four-wheel drive variant. The rear axle reduction ratio shall be the same on the two-wheel drive and the four-wheel drive FLTT variants. The rear axle shall have a limited-slip differential and a reduction ratio equal to or greater than 3.73:1. The rear axle shall feature single rear wheels. A “special option” from the OE chassis-cab manufacturer may be required to meet this axle ratio specification. Non-directional mud and snow tires shall be provided on the Class 2 FLTT. The two-wheel drive and four-wheel drive FLTT shall have a maximum wall-to-wall turning radius of 21 feet or less. The wheelbase shall be between 90 and 100 inches. The balance of these two parameters shall be optimized.

3.6.4 Brakes.

3.6.4.1 Parking brake. The FLTT shall be equipped with a parking brake that shall be in compliance with SAE J293. The FLTT manufacturer shall ensure that any parking-brake cable relocation or cable shortening (resulting from modifying/shortening the wheelbase of the commercial truck chassis) is equivalent to the OE chassis-cab manufacturer’s design in providing parking brake cable tension.

3.6.4.2 Service brakes. The FLTT shall feature a braking system in compliance with Federal Motor Vehicle Safety Standards (FMVSS) for vehicles of this GVWR at the time of manufacture.

3.6.5 Pintle hooks. The FLTT shall be equipped with four pintle hooks that incorporate a cotter or snapper pin as a secondary-locking device, and conform to CID A-A-52550, Type 1, Class 2. Pintle hooks shall be attached to the pintle hook mounting structure with a minimum of four ½-13 or ½-20, Grade 8 fasteners. The pintle hook fasteners shall be installed with a design fastening torque applied such that an adequate preload is introduced in the fastener to prevent losing or fatiguing the fasteners and/or the pintle hook.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.6.5.1 Front pintle hook position. The front of the FLTT shall be equipped with one center-mounted pintle hook, with the coupling position of the pintle hook located at 18.5 inches from the ground, plus or minus 0.5 inch. The front pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall comply with the following:

a. The rated load shall be the combination of load forces that the pintle hook assembly and structure must support or resist in a static state. At a minimum, the rated load shall be represented by a force value of 6,000 pounds applied in the same vector directions as DBPF.

b. The dynamic load shall be the rated load with adjustments for loads and accelerations in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate and demate, load and unload operations.

c. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, whichever is greater. This design load shall be considered the minimum load for attaining the design stress levels.

3.6.5.2 Rear-outboard pintle hook positions. The rear of the FLTT shall be equipped with two outboard pintle hooks located in the same vertical plane, with the pintle hook centerlines at 43.5 inches, plus or minus 0.5 inch, to each side of the FLTT centerline, with the coupling positions of the pintle hooks located at 18.50 inches, plus or minus 0.5 inch, from the ground. The rear-outboard pintle hook assemblies and mounting structures shall comply with the following:

a. The rated load shall be the combination of load forces that the pintle hook assembly and structure must support or resist in a static state. At a minimum, the rated load shall be represented by a force value of 6,000 pounds applied in the same vector directions as DBPF.

b. The design load shall be the rated load with adjustments for loads and accelerations in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate and demate, load and unload operations.

3.6.5.3 Rear-center pintle hook position. The rear of the FLTT shall be equipped with one center-mounted, extendable pintle hook with pintle hook coupling positions that are height adjustable from 12 inches to 18.5 inches from the ground. A minimum of four evenly spaced coupling positions (including 12 inches and 18.5 inches) shall be provided. Height adjustment shall be easily made by the operator without the use of tools. The rear-center extendable pintle hook shall not extend more than 10.5 inches beyond the rear plane of the truck when retracted for towing. The extendable unit shall consist of a self-locking sliding bar that shall allow the hook backward movement of not less than 10.5 inches, or more than 12 inches, and a lateral swing movement of 45 degrees in either direction from the centerline of the FLTT. The unit shall be fitted with rollers or low-friction pads, and provisions shall be made for lubricating all moving parts. The normal towing position shall be with the pintle hook/hitch assembly in the retracted and straight position, rather than the extended and/or swung position. The extendable bar shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

be equipped with a safety grab handle, attached to the pintle hook/hitch assembly, and positioned above the pintle hook, to allow for easy user access and function of the extendable hitch's front-to-rear and side-to-side hitch motion, as well as the pintle hook height adjustments. The extendable bar shall be easily removable to facilitate inspection and maintenance. Once the lunette or towing eye of the towed load is secured, the extendable unit shall be engaged in the closed and straight positions for towing by the backward movement of the FLTT until the self-locking device is secure. Any shear pins, locking pins, or related pintle hook components that are designed to be removed for adjustments by the user shall be attached to the FLTT with lanyards. Removable items that are heavier (such as the pintle hook assembly, if applicable) shall be attached with an appropriate, heavier gauge lanyard. Shear pins that use a locking pin for retention shall utilize a locking pin that is a double-loop hair-pin design. If constructed of mild steel, the hitch shall have hardened steel bushings and pins at all pivot and locking points. The bushings shall be easily removed for service. The rear-center pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall comply with the following:

In the extended (open) position:

a. The rated load shall be the combination of load forces that the pintle hook assembly and structure must support or resist in a static state. At a minimum, the rated load shall be represented by a force value of 6,000 pounds applied in the same vector directions as DBPF.

b. The design load shall be the rated load with adjustments for shock loads associated with mate and demate, load and unload operations.

In the retracted (closed) position:

a. The rated load shall be the combination of load forces that the pintle hook assembly and structure must support or resist in a static state. At a minimum, the rated load shall be represented by a force value of 6,000 pounds applied in the same vector directions as DBPF.

b. The dynamic load shall be the rated load with adjustments for loads and accelerations in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate and demate, load and unload operations.

c. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, whichever is greater. This design load shall be considered the minimum load for attaining the design stress levels.

3.7 Engine and related equipment.

3.7.1 Engine. The FLTT shall have a commercially available, liquid-cooled, diesel engine of not less than six cylinders and power train sufficient to develop the required drawbar pull and obtain

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

the required FLTT speeds. The diesel engine shall operate on high-sulfur diesel fuels (containing over 15 parts per million (ppm) sulfur), and ultra-low sulfur diesel fuels (containing less than 15 parts per million (ppm) sulfur). At a minimum, the FLTT shall meet a Tier III level exhaust and evaporative emissions standard (see 7.4.1) for on-highway engines. The engine shall include an anti-restart, as well as a neutral safety start feature. Engine shall have an automatic engine shutdown system for high coolant temperature, low oil pressure, and engine over speed (engine in excess of 110% of rated speed).

3.7.2 Engine starting aids. The engine shall start within 15 seconds cranking in any ambient temperature within the required operating range of the FLTT. Internal engine starting aids, fluid starting aids, and heat from the winterization system (see 3.7.6) may be used prior to and during the start period to facilitate engine starting under the following conditions in Table V:

TABLE V. Engine starting aids.

Temperature Range	Starting Aids Permitted
40 °F through 125 °F	None
0 °F through 39 °F	Internal engine starting aids and fluid starting aids
-40 °F through -1 °F	Internal engine starting aids, fluid starting aids, and heat from the winterization system

3.7.3 Engine cooling system. The FLTT engine shall possess sufficient cooling capacity to perform continuous maximum performance towing in high ambient temperatures of 125 ^{o}F without exceeding the engine OEM's maximum allowable temperatures on the individual engine components.

3.7.4 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. The exhaust system outlet shall exit on the right-hand side, over the rear axle and behind the rear wheel, oriented at 90 degrees from the centerline of the FLTT, and not directed toward the ground.

3.7.5 Transmission. Transmission shall be of the automatic type, with an integral, limited-slip, inter-axle differential. Maximum allowable transmission temperature during all operations shall not exceed 260 ^{o}F.

3.7.6 Winterization system. A type C winterization system shall be provided for starting in temperatures down to -20 ^{o}F. When specified (see 7.2), a heavy-duty (type A) winterization package adequate for storage down to -65 ^{o}F and operation to -40 ^{o}F. The winterization systems may include heaters for engine coolant, engine oil, and the fuel tank, as well as battery warmers. The winterization system shall be designed to operate from an external 110 volt AC, 60 Hz power source utilizing the external electrical connections (see 3.7.6.1). The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.7.6.1 Winterization plugs and connectors. Winterization package components with external power requirements shall be a covered, 2 pole, 3-wire grounding male inlet plug using 120VAC/20A circuitry, in accordance with National Electrical Manufacturers Association (NEMA) WD 6.

3.8 Electrical system. The FLTT shall be equipped with a 12V, negative ground electrical system. All external chassis wiring shall incorporate weather-proof connections and terminations. Exceptions to this are under-hood terminal strips and electrical posts that are protected from the elements. Any wire splicing (typically resulting from special-request options) shall incorporate a mechanically-crimped connector, solder, and heat-shrink tubing. Crimped-on wire connectors shall be installed with a corresponding crimping tool. All wiring shall be sized to handle the system voltage and current levels. Battery cables shall be identified with "+" and "-" or red and black markings, and shall not be spliced.

3.8.1 12-volt trailer light harness receptacle. The rear of the FLTT shall be equipped with an SAE J560 electrical receptacle, provided on the FLTT's centerline, within 12 inches (vertical distance) above the top of the center pintle hook while in its uppermost height position, and within 15 inches forward of the base of the pintle hook. All terminals of the receptacle shall be wired in accordance with SAE J560. The FLTT shall be equipped with an aftermarket, heavy-duty trailer-light circuit protection system that shall isolate the trailer's lighting circuits from the FLTT's electrical system. This system shall power any trailer lights directly from the FLTT's battery, with a minimum system circuit protection of 20 amps. The #7 terminal (blue wire) in the J560 connector shall be an auxiliary, ignition-hot circuit, also capable of handling 20 amps.

3.8.2 Lighting. The FLTT lighting shall be equivalent to the standard commercial offering to include a rear adjustable aim floodlight, parking lights, head lights (high and low beam), brake lights, turn signals, hazard lights, and reverse lights. Basic lighting system shall be in compliance with FMVSS requirements. A flashing amber beacon light shall be mounted above the roofline on the FLTT's centerline. A switch shall be mounted inside the cabin within reach of the seated operator for actuating the beacon light. Two double-faced signal lamps (SAE J588) shall be mounted above the roofline, approximately 48 inches apart and centered on the FLTT's centerline, and shall be integrated with the FLTT's taillights. The flashing beacon light and double-faced signal lamps shall be mounted on a light bar that is not fastened to the cab that runs approximately 1 inch above the roofline, to avoid drilling and cutting into the cab. The rear adjustable aim floodlight, of not less than 3000 candlepower, shall be mounted on the FLTT so that the rear working area behind the cab and the rear coupling positions are illuminated. The FLTT shall be equipped with an adjustable spotlight, capable of illuminating 180 degrees of elevation and 360 degrees of rotation, and located on the driver's side of the cab. The spotlight shall be installed in the driver's door-to-windshield pillar, and steerable from the driver's position, inside the cab.

3.8.3 Batteries and battery compartment.

3.8.3.1 Batteries. Batteries shall be of the commercial maintenance-free type that come standard with the commercial chassis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.8.3.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

3.9 Breakaway cable hookup. The rear of the FLTT shall be equipped with an emergency trailer-breakaway cable attachment point, capable of mating with a National Aerospace Standards (NAS) 1281C10 spring snap hook. The attachment point shall be located on the FLTT's centerline, within 8 inches (vertical distance) above the top of the center pintle hook while in its uppermost height position, and within 4inches forward of the base of the pintle hook. The attachment point shall not be installed on the pintle hook assembly.

3.10 Safety chain attachment points. The rear of the FLTT shall have two safety chain attachment points for use when towing, composed of hinged D-rings suspended in the downward position. The D-rings shall be located symmetrically on each side of the center pintle hook at a distance of 26 inches above the ground. The vertical centerline of the D-rings shall be located at a parallel distance of not greater than 6 inches to the left and right of the FLTT's centerline. The rear edge of the D-rings shall be within 2 inches forward of the base of the pintle hook. The D-rings shall not be installed on the pintle hook assembly. Each D-ring shall be capable of withstanding a maximum shock load of 6,000 pounds and shall have the ability for standard safety chain hooks to be attached. The inside radius of the safety chain hook is between 1 and 3 inches.

3.11 Chassis modifications.

3.11.1 Chassis frame modifications. The FLTT shall be based on a commercial truck chassis also known as a chassis cab. The commercial truck chassis cab shall be modified to the specified wheelbase by relocating the rear axle, suspension, and other affected components forward on the original chassis frame rails. Back-halved chassis or spliced-in rear sub-frames, creating a joint in the center of the chassis length, shall not be used in the modification. Chassis components shall be relocated and/or modified as necessary to avoid excessive loading, suboptimal drive angles, and driveline vibrations. Modifications to chassis components, especially load bearing suspension components, shall be performed using an engineering approach and with quality workmanship so as to avoid weakening the structural integrity of the component. Excess length at the rear of the chassis frame rails from the forward relocation of chassis components shall be removed using a process that leaves a smooth cut finish. Raw metal exposed by the modifications shall be primed and painted chassis color. The relocation of chassis components shall not result in improperly loaded fasteners.

3.11.2 Chassis fuel system modifications. Modifications to the truck chassis shall not result in exposed fuel supply or fill lines. During the wheelbase-modification process, the FLTT manufacturer may determine that splicing into the OEM metal fuel lines with flexible fuel-lines is necessary (such as when routing flexible fuel lines to a new fuel tank location). In that event, fuel injection-rated hoses shall be utilized, as well as fuel injection-rated clamps, where clamps are applicable. If modification of the OEM metal fuel lines is necessary, it shall be done so as to maintain OEM integrity to prevent fuel lines from leaking or blowing off under pressure. Flexible fuel-line connections can be accomplished to the ends of the metal lines that have been

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

cut and flared after a tube nut has been placed over the end of the cut metal line. The flared metal line and integral tube nut shall then be threaded into a female-inverted, “field-attachable” brass or steel fitting and tightened. This type of application does not necessarily require a hose clamp when the appropriate working pressures are observed, and the correct flexible hoses for the application are selected.

3.12 Workmanship. The FLTT, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.12.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.12.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.12.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.12.4 Cleaning. The FLTT shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production FLTT shall be subjected to the analyses, demonstrations, examinations, and tests described in 5.5.1 through 5.5.14. The contractor shall provide or arrange for all test equipment and facilities.

5.3 Conformance inspection. Each production FLTT shall be subjected to the examination described in 5.5.1.

5.4 Inspection requirements.

5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

5.4.3 Test rejection criteria. Throughout all tests specified herein, the FLTT shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer's (OEM's) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the FLTT components or between the FLTT, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

j. Shutdown faults from:

    i. Engine cooling system.
    ii. Engine lubrication system.
    iii. Engine protective circuits.

5.5 Test methods.

5.5.1 Examination of product. Each FLTT shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 5.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each FLTT function shall be verified. Certifications and analyses shall be provided in accordance with Table VI. Each production FLTT shall be inspected to a Government-approved reduced version of the checklist.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

TABLE VI. Certifications and analyses.

Paragraph	Required Certifications and Analyses
3.2 Design and construction.	Contractor certification that the FLTT is in accordance with all requirements of SAE specifications ARP1247 and AIR 1363.
3.2.9.2 Cab visibility.	Contractor certification that the cab visibility requirements of 3.2.9.2 in accordance with MIL-STD-1472 are met or exceeded.
3.2.10 Tires and wheels.	Contractor certification that the tire and wheel selections are in accordance with the *Tire and Rim Association*.
3.2.11 Air system.	Contractor air system analysis (see 5.5.8).
3.5.1 Air transportability.	Contractor air transportability analysis (see 5.5.7.1).
3.5.1.1 Shoring.	
3.5.2 Tie downs.	Contractor tie down provision analysis (see 5.5.7.3.1).
3.5.3 Lifting provisions.	Contractor lifting provision analysis (see 5.5.7.4.1).
3.6.4.1 Parking brake.	Contractor analysis demonstrating equivalence of the OEM and modified parking brake (see 5.5.11).
3.6.5 Pintle hooks. (including all subparagraphs)	Contractor analysis of pintle hook requirements (see 5.5.12).
3.7.1 Engine.	Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels. Engine manufacturer application approval for the engine and its installation, including cooling system, lubrication system, and mounting system.
3.7.3 Engine cooling system.	
3.7.5 Transmission.	Transmission manufacturer certification that the transmission is in accordance with all requirements of 3.7.5.
3.10 Safety chain attachment points.	Contractor analysis of the safety chain attachment point (see 5.5.13).

5.5.3 Electromagnetic interference test. A first production FLTT shall be tested in accordance with MIL-STD-461: RE 102 and RS 103 to demonstrate compliance with 3.2.5.

5.5.4 Noise level test. The noise level at the operator's ear with the engine operating at maximum rpm and windows and doors closed shall be measured to demonstrate compliance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.2.9.3. If noise level is between 84 and 94 dB(A), the noise warning plate required by 3.2.9.3 shall be on all FLTTs delivered on this contract.

5.5.5 Environmental testing.

5.5.5.1 High temperature storage and operation test. A first production FLTT shall be tested in accordance with MIL-STD-810, Method 501.6, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.3.1 and 3.3.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.5.2 Low temperature storage and operation test. A first production FLTT shall be tested in accordance with MIL-STD-810, Method 502.6, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.3.1 and 3.3.2, as well as the engine starting requirement of 3.7.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

5.5.6 Weight and dimension tests.

5.5.6.1 Weight and center of gravity test. The weight and center of gravity of a first production FLTT shall be measured to demonstrate compliance with the center of gravity requirement of 3.2.2 and the weight requirement of 3.4.1.

5.5.6.2 Dimension measurement. A first production FLTT shall be measured to demonstrate compliance with the dimensional requirements of 3.4.2, and the wheelbase requirement of 3.6.3.

5.5.7 Transportability verification.

5.5.7.1 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 3.5.1, 3.5.1.1, and 3.5.1.3. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.5.1. The evaluation shall also include a dimensional analysis for the FLTT while traversing the ramp and while loaded aboard C-130, C-17, and C-5 aircraft.

5.5.7.2 Equipment removal and reconfiguration demonstration. A first production FLTT shall be configured for transport on C-130, C-17, and C-5 aircraft and then reconfigured for operation to demonstrate compliance with 3.5.1.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

5.5.7.3 Tie down provision verification.

5.5.7.3.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 3.5.2.

5.5.7.3.2 Tie down provision test. A first production FLTT shall be tested to demonstrate compliance with the tie down provision requirements of 3.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

5.5.7.4 Lifting provision verification.

5.5.7.4.1 Lifting provision analysis. An engineering analysis shall be performed to demonstrate compliance with the lifting provision requirements of 3.5.3.

5.5.7.4.2 Lifting provision test. A first production FLTT shall be tested to demonstrate compliance with the lifting provision requirements of 3.5.3.

5.5.8 Air system analysis. A schematic of the air system plumbing shall be submitted to the government representative at first production verification. This schematic shall show the location of the air system components on the FLTT and shall show that all components required by 3.2.11 are provided.

5.5.9 Drawbar pull test. The first production FLTT shall be tested for verification of the DBPF on dry level concrete using a load sensing device and a fixed anchoring position. The drawbar position shall be horizontal and level to demonstrate compliance with 3.6.2.

5.5.10 Driveline and steering. The first production FLTT shall be demonstrated with a test drive by the government representative. The FLTT shall also be tested for verification of the maximum wall-to-wall turning radius specified in 3.6.3. The fully ballasted first production FLTT shall be tested as follows to demonstrate compliance with the turning radius specification:

a. Facility. Perform the test on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the FLTT being tested.

b. Equipment required.

    i. A device suitable for measuring three times the length of the FLTT being tested with an accuracy of at least ±1 inch.
    ii. Markers or marking device suitable for marking the pavement.
    iii. A plumb bob suitable for locating a point on the pavement directly below a fixed point on the FLTT.
    iv. A test report notebook or similar record forms to be used as a test report work sheet and incorporated into the documentation package.

c. Test Conditions.

    i. The FLTT shall be fully ballasted and in operational form.
    ii. The FLTT steering system shall be fully operational, and the steering linkage stops shall be adjusted to the manufacturer's specified production tolerance limits.

d. Test Procedure.

    i. The FLTT shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

ii. Continue driving the slow full cramp circle.
iii. At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the FLTT using the service brakes.
iv. At each stop, place a plumb bob against the outermost point of the FLTT and mark the spot on the ground directly below where the plumb bob comes to rest.
v. Measure and record the straight line distances between each pair of points, that is, Lengths AB, BC, and CA.
vi. Calculate the wall-to-wall turning radius (R) as follows:

$R = D / 2$ where

$$D = \frac{(AB \times BC \times CA)}{2 \times \sqrt{(S \times (S - AB) \times (S - BC) \times (S - CA))}} \text{ and}$$

$$S = (AB + BC + CA) / 2$$

e. Repeat steps (i) through (vi) with the FLTT moving in the opposite direction.

5.5.11 Parking brake analysis. The first production FLTT shall be inspected for verification of a functional, OEM parking brake as required by 3.6.4.1. If applicable, documentation of any "before" and "after" parking-brake cable relocations, cable-shortening modifications, or associated modification processes shall be presented to the government representative at the first production verification. This documentation shall be verification that the OE chassis-cab manufacturer's design and the FLTT manufacturer's design modifications are equivalent when producing parking-brake cable tension. If applicable, the documentation should demonstrate equivalence between the original and modified cable mounting locations verses the original and modified cable lengths.

5.5.12 Pintle hook analysis. A strength analysis report shall be submitted to the Government representative at first production verification that contains supporting calculations to show that the front and rear pintle hooks and mounting structures are in compliance with the respective strength specifications of 3.6.5 and all subparagraphs. The analysis shall include methodology leading to the arrival of values for static and dynamic loading, and ultimately the design load. Additionally, the rear coupler and mounting structure calculations shall be supplemented with a Finite Element Analysis (FEA) to further support compliance with the strength specification.

5.5.13 Safety chain attachment point analysis. A strength analysis of the safety chain attachment points and mounting (through ratings and/or calculations) shall be submitted to the Government representative at first production verification of the air system-equipped FLTT to demonstrate compliance with 3.10.

5.5.14 Mobility test. The first production FLTT shall be tested for compliance with the requirements of 3.6.1. The first production FLTT shall be tested for verification of the forward and reverse speeds on a flat asphalt or concrete surface without a trailing load, using the FLTT's speedometer as the speed-measuring device. The FLTT shall be tested for verification of its capability to tow a 40,000 pound trailer on a 2.5 % grade at a minimum of 10 MPH on an asphalt

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

or concrete surface, using the FLTT's speedometer as the speed measuring device. The FLTT shall be tested for verification of its capability to operate on 10 percent side slopes and 12 percent longitudinal grades without fluid spillage.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES.

7.1 Source of documents.

7.1.1 Department of Defense and Federal documents. Department of Defense and Federal documents, except for GOST 10227-86, are available online at http://quicksearch.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094. A copy of GOST 10227-86 can be obtained from the Procuring Contracting Officer (PCO).

7.1.2 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far/.

7.1.3 ASTM documents. Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959 or online at http://www.astm.org.

7.1.4 AWS documents. AWS Standards may be obtained at http://www.anweld.org/ or from the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126.

7.1.5 SAE documents. SAE Standards may be obtained at http://www.sae.org or from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

7.1.6 NAS documents. Application for copies should be addressed to Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington VA 22209 or online at http://www.aia-aerospace.org/national_aerospace_standards/ .

7.1.7 NEMA documents. Application for copies should be addressed to NEMA, 1300 North 17th Street, Suite 1847, Rosslyn VA 22209, http://www.nema.org/ .

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. Classification (see 2.1)

c. Temporary CPC (see 3.2.1.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

d. Air system (see 3.2.9.1, 3.2.11)

e. Winterization A (see 3.3.1, 3.7.6)

f. Packaging requirements (see 6.0)

7.3 Definitions.

7.3.1 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

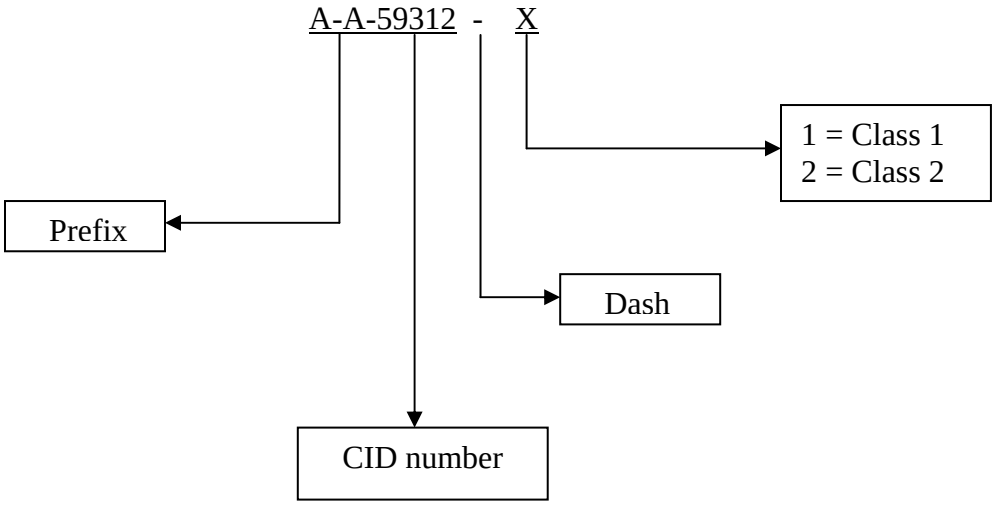
7.3.2 Special tool. A tool that is not commercially and readily available from a source other than the FLTT contractor.

7.4 Additional information.

7.4.1 Engine waiver. Tier III represents the highest level of exhaust and evaporative emissions standards that will permit the use of high sulfur fuels. The Air Force possesses an Environmental Protection Agency (EPA) waiver to allow for deviation from EPA emissions guidelines. The waiver will be transferred from the Air Force to the engine manufacturer. The transfer process requires the engine manufacturer to supply a decal for the engine stating "insert engine description here – This engine has an exemption for national security under 40 CFR 89.908 and/or 40 CFR 1068.225".

7.4.2 Part or Identification Number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-59312:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

7.4.3 National Stock Number (NSNs). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

Master NSN	Classification
1740-01-068-8945YW	Two-Wheel Drive Tow Tractor
1740-01-452-4117YW	Four-Wheel Drive Tow Tractor

7.5 Key words.

Aircraft ground equipment – (AGE)
Aircraft parking
Cortec VpCI 389
Ground handling
Flight line, military aircraft
FLTT
Munitions handling
Towing, military aircraft and trailer

7.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST:

Custodians:
Army - AV
Air Force - 84

Preparing Activity:
Air Force - 84

Reviewer:
Air Force - 99

Agent:
Air Force – 99
(Project 1740-2015-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.